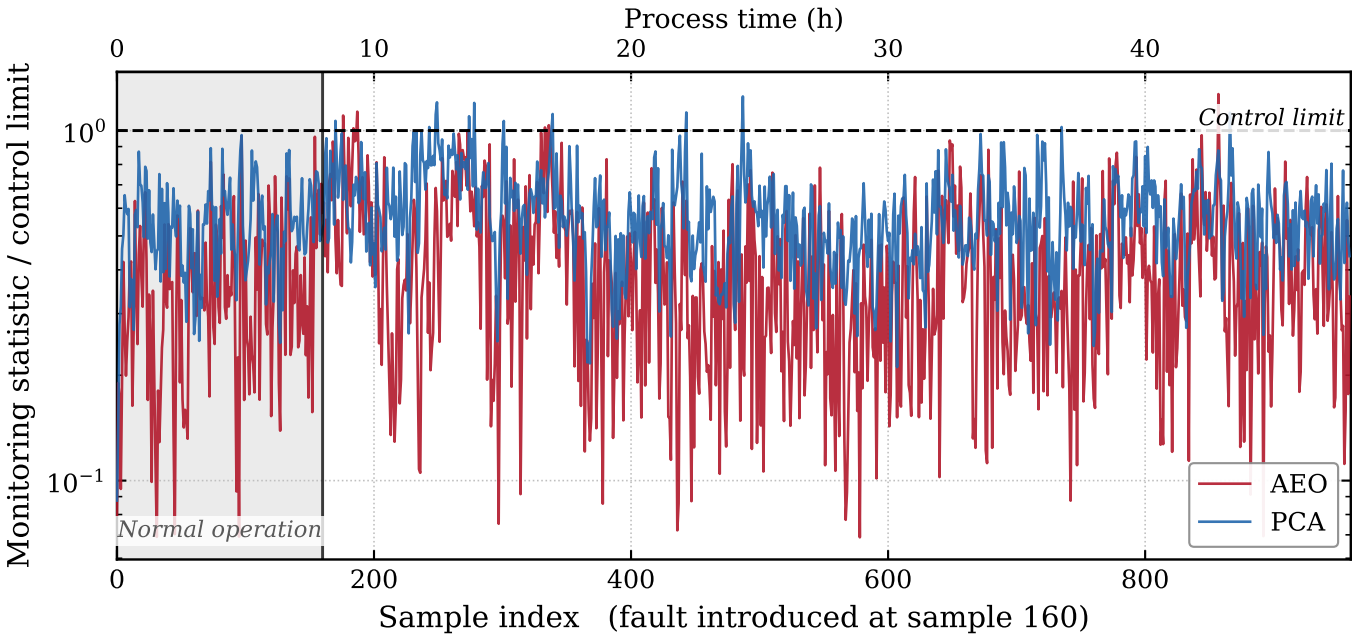
Fault Detection Performance for IDV(3)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	168.0 samples (504 min)	MDR 0.988	detected in 2/7 runs
PCA	delay	248.5 samples (746 min)	MDR 0.989	detected in 2/7 runs

Fault not reliably detected by either method (caught in fewer than half the runs)